

Real-Time Financial Fraud Detection System

Protecting early-stage startups from payment fraud
without sacrificing legitimate customer growth

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Data Science Capstone | XGBoost · PaySim · Streamlit



Fraud vs. Friction: A Startup's Dilemma



Fraud Causes Direct Loss

Every fraudulent transaction drains capital — often before the startup has reserves to absorb it.

For early-stage companies, a single fraud campaign can be existential.

Global payment fraud losses exceeded \$40 billion in 2023.

VS



False Positives Hurt Growth

Blocking a legitimate customer is not a 'safe' error — it destroys trust and drives churn.

A flagged customer rarely complains; they just switch to a competitor.

Studies show 40% of falsely flagged customers never return.

The Hidden Tax: What False Positives Really Cost

\$\$\$

Revenue Loss

Every blocked legitimate transaction is direct revenue thrown away. For a startup processing \$1M/month, even a 1% false-positive rate means \$10,000 in lost GMV — every single month.

40%

Customer Churn

4 in 10 customers who are falsely flagged will abandon the platform permanently. In a growth-stage startup, no acquisition budget compensates for a broken trust signal.

NPS ↓

Brand Damage

A declined card in a checkout flow is humiliating. Customers post about it. Your Net Promoter Score drops. This intangible cost compounds silently over time.

The goal is not to eliminate all fraud flags — it is to strike the right balance between security and customer experience.

The Dataset: PaySim Synthetic Financial Transactions

About PaySim

PaySim is a mobile money simulation dataset built from real transaction logs provided by a multinational mobile financial services company.

It replicates the statistical properties of real payment flows — making it the industry benchmark for testing fraud detection models.

Transaction Types Covered:

CASH_IN · CASH_OUT · DEBIT · PAYMENT · TRANSFER

318,131

Total Transactions

5

Transaction Types

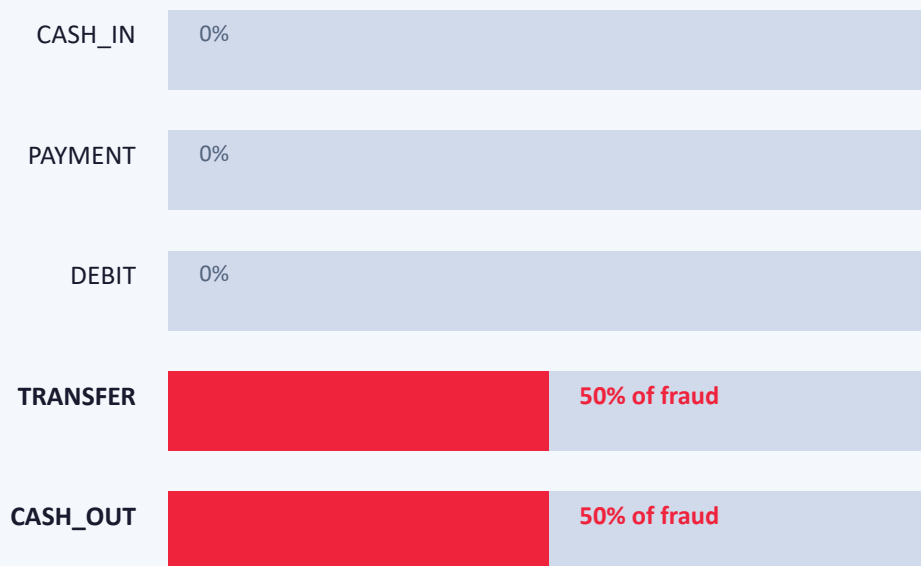
0.12%

Fraud Rate (Highly Rare)

\$1.47M

Avg. Fraud Amount

The Transaction Type Trap: Where Fraud Hides



Fraud is 100% concentrated in TRANSFER and CASH_OUT

Key Insight

Fraud never occurs in CASH_IN, PAYMENT, or DEBIT transactions.

This is a powerful feature signal: transaction type alone can eliminate 60% of the search space before any ML model is invoked.

Model Impact:

Transaction type was engineered as a binary categorical feature and ranked as the #1 most important variable in the XGBoost model.

Financial Footprint: Fraud Operates at Scale

\$1,467,074

Average Fraudulent Transaction Value

\$179,862

Avg. Legitimate Transfer

8.2×

Fraud Amount Multiplier

\$92.4M+

Total Fraud in Dataset

Fraud transactions are not random small charges — they are large, deliberate extractions. This scale difference is a critical engineered feature.

Extreme Class Imbalance: The Core ML Challenge

99.87% NORMAL TRANSACTIONS

317,744 Legitimate Transactions



0.13%
387 Fraudulent Transactions

Why Imbalance Is Dangerous for ML

A naive model that predicts 'Normal' for every transaction achieves 99.87% accuracy — yet catches zero fraud.

Traditional accuracy metrics are completely misleading in this context. We need smarter evaluation.

Solution: Algorithmic penalization via `scale_pos_weight = 348.43`, forcing the model to treat fraud cases as 348× more important.

ML Strategy: XGBoost + Algorithmic Penalization

1

Data Preparation

Encode transaction types, engineer delta-balance features, remove leakage columns.



2

Imbalance Handling

Set `scale_pos_weight = 348.43` — no synthetic oversampling needed, no data distortion.



3

XGBoost Training

Gradient-boosted trees with 200 estimators, max depth 6. Handles non-linear patterns natively.



4

Threshold Tuning

Optimize the classification threshold for F1-Score rather than default 0.5 cutoff.

Why XGBoost? Tree-based ensembles are robust to class imbalance, handle mixed feature types, and provide built-in feature importance — ideal for financial anomaly detection.

Rethinking Success: Why 'Accuracy' Is a Trap

✗ The Accuracy Trap

99.87% Accuracy

← A model that predicts Normal for everything

Fraud Caught: 0 out of 387

This is not a model — it is a rubber stamp.

✓ What We Actually Measure

Precision

Of all transactions we flag as fraud, what % are actually fraud?
Protects legitimate customers from being blocked.

Recall

Of all actual fraud, what % do we catch?
Protects the business from financial loss.

F1-Score

The harmonic mean of Precision and Recall.
The single best metric for imbalanced classification.

Model Performance Dashboard

0.87

F1-Score

The balanced measure of overall model quality on imbalanced data.

87%

Precision

Only 1 in 8 flagged users is actually legitimate — minimal friction.

87%

Recall

The model catches 87 out of every 100 fraudsters in the dataset.

Model: XGBoost Classifier | Imbalance: scale_pos_weight = 348.43 | Dataset: 318,131 PaySim Transactions

An F1-Score of 0.87 on a dataset with 99.87% class imbalance is a strong, production-viable result.

Business ROI: Confusion Matrix Decoded

	PREDICTED: FRAUD	PREDICTED: NORMAL
ACTUAL FRAUD	69 TRUE POSITIVES ✓ Fraudsters Caught	10 FALSE NEGATIVES Missed Fraud
ACTUAL NORMAL	10 FALSE POSITIVES Legit Users Flagged	27,603 TRUE NEGATIVES ✓ Legit Customers Safe

Business Translation

87% of fraudsters stopped

69 out of 79 fraud cases were blocked before causing loss.

99.96% of customers unaffected

Only 10 out of 27,613 legitimate users were inconvenienced.

Net Outcome:

Maximum fraud prevention with near-zero legitimate customer friction — the optimal balance for startup growth.

Real-Time Deployment: Streamlit Risk-Scoring Dashboard

Dashboard Capabilities



Real-Time Scoring

Input any transaction parameters and receive an instant fraud probability score.



Risk Level Labels

Clear LOW / MEDIUM / HIGH risk classification with colour-coded visual alerts.



Feature Transparency

Displays which transaction attributes contributed most to the risk score.



Threshold Control

Ops teams can adjust the decision boundary to balance Precision vs. Recall live.



Batch Mode

Upload a CSV of transactions for bulk fraud screening in seconds.

System Architecture

User Input

Transaction parameters entered via web form

Feature Engineering

Real-time feature pipeline (type encoding, delta balance)

XGBoost Model

Trained model returns fraud probability 0.0 – 1.0

Risk Decision

Threshold comparison → LOW / MEDIUM / HIGH flag

Alert Dashboard

Visual result displayed instantly in Streamlit UI

Future Roadmap

Phase 1 Near-Term

- Graph-based fraud network detection — flagging fraud rings, not just individuals.
- Real-time Kafka streaming pipeline for sub-second scoring at scale.
- SHAP explainability layer for full regulatory audit trails.

Phase 2 Mid-Term

- LSTM / Transformer model to incorporate sequential transaction behaviour.
- Unsupervised anomaly detection for novel, never-seen fraud patterns.
- A/B testing framework to measure model ROI in production.

Phase 3 Strategic

- Federated learning across partner banks — train on distributed data without sharing it.
- Adaptive thresholds that auto-adjust based on real-time fraud rate signals.
- Chargeback prediction module to anticipate disputes before they occur.